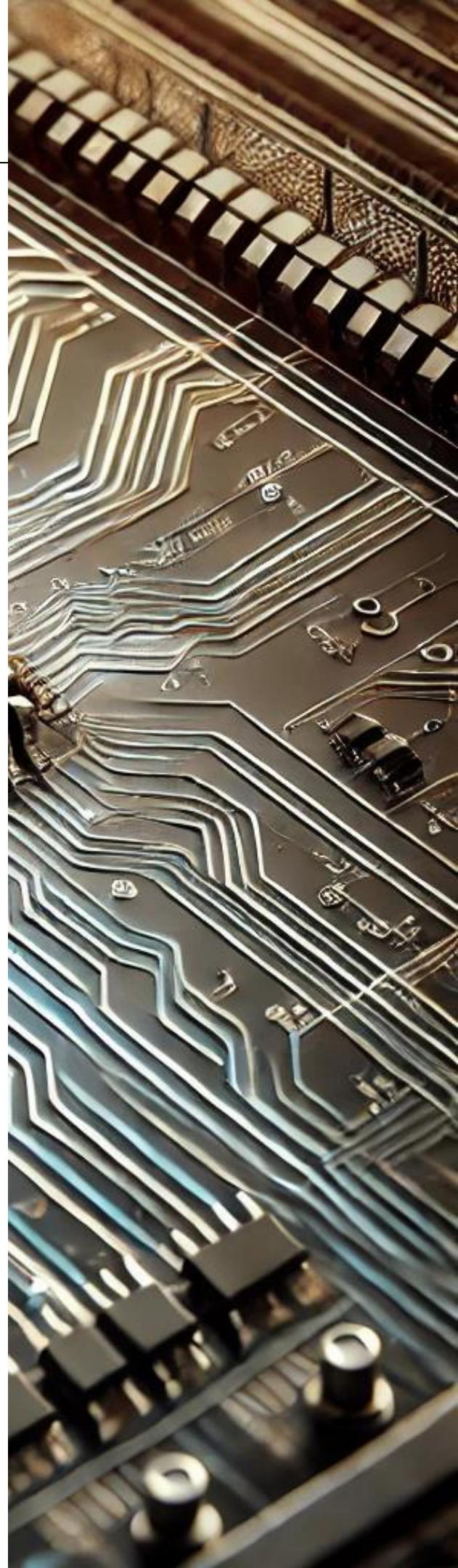


10주차 MSI/LSI 연산회로

4-bit Parallel Adder/Subtractor
BCD Adder/Subtractor

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Part 1

4 bit binary

Parallel Adder

/ Subtractor

1-1

Binary Adder

1-2

4-bit Adder

1-3

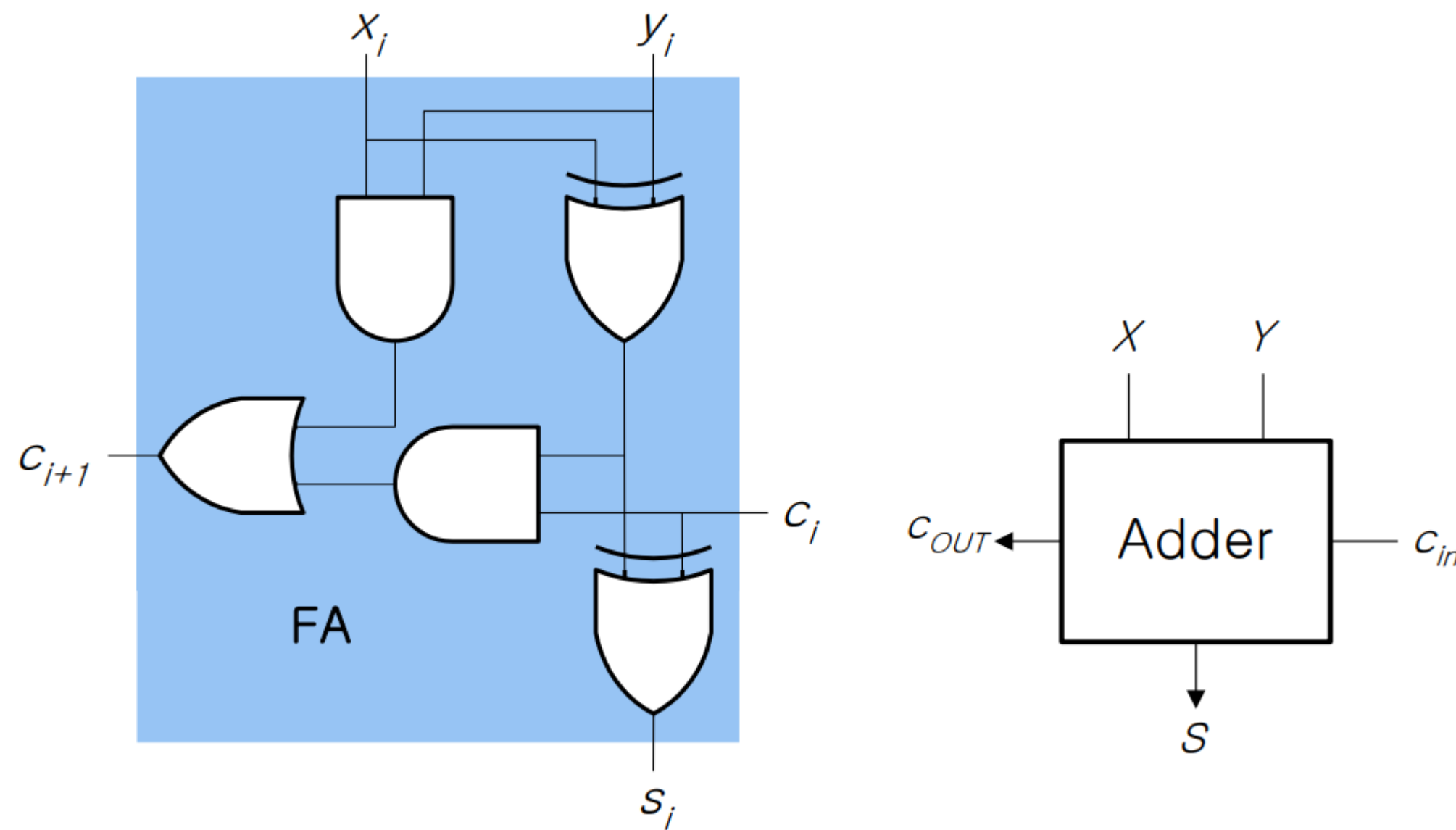
4-bit Addder 회로

1-4

Subtractor

Binary Adder

4-bit Binary Parallel Adder/Subtractor는 이진수의 덧셈과 뺄셈을 동시에 수행할 수 있는 디지털 회로입니다.
이 회로는 두 개의 4비트 이진수 A와 B를 입력받아 그 합 또는 차를 출력합니다.



1-bit binary Adder

$$S_i = X_i \oplus Y_i \oplus C_i$$

$$C_{i+1} = X_i Y_i + (X_i \oplus Y_i) C_i$$

위 식을 확장하여 4-bit binary Adder을 만들 수 있습니다.

4-bin Adder

▪ $x_{i+3}x_{i+2}x_{i+1}x_i + y_{i+3}y_{i+2}y_{i+1}y_i$ (input carry c_i)

- carry-generate function $g_i = x_i y_i$

- carry-propagate function $p_i = x_i \oplus y_i$

- $c_{i+1} = g_i + p_i c_i$ $c_{i+2} = g_{i+1} + p_{i+1} c_{i+1}$

- $c_{i+3} = g_{i+2} + p_{i+2} c_{i+2}$ $c_{i+4} = g_{i+3} + p_{i+3} c_{i+3}$

▪ Express in terms of c_i

- $c_{i+1} = g_i + p_i c_i$

- $c_{i+2} = g_{i+1} + p_{i+1} g_i + p_{i+1} p_i c_i$

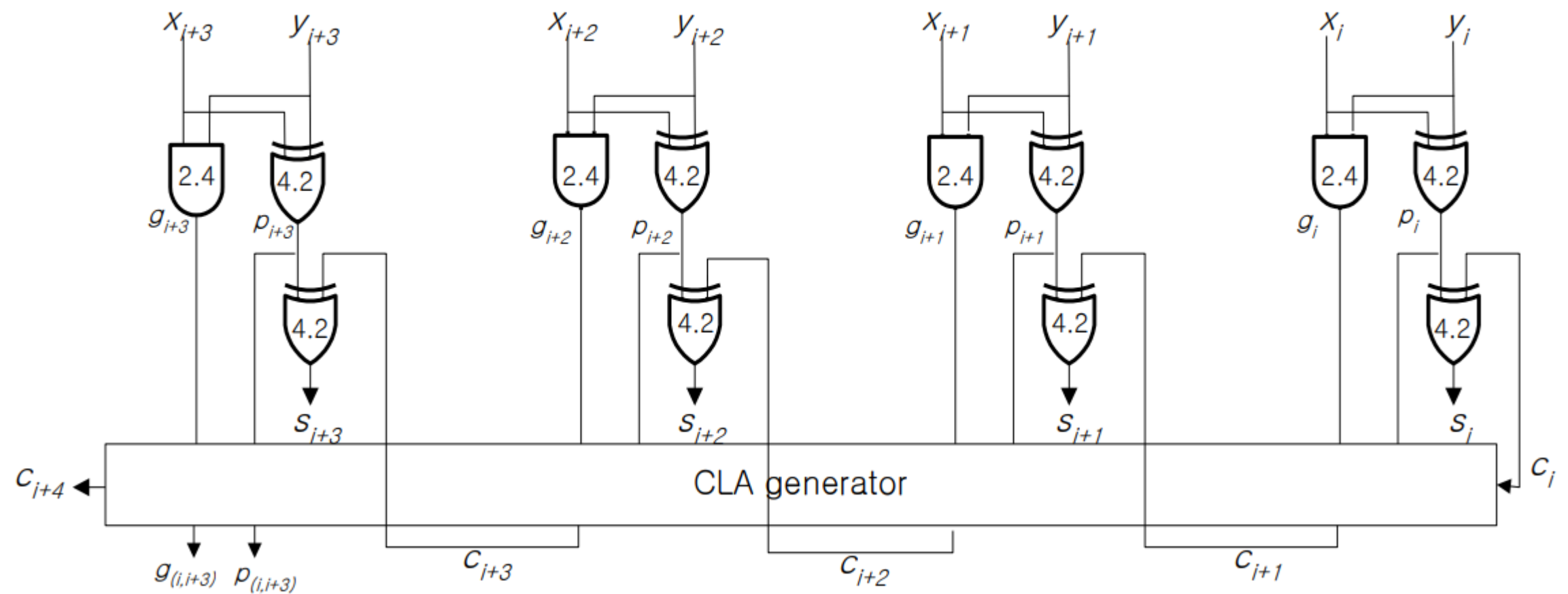
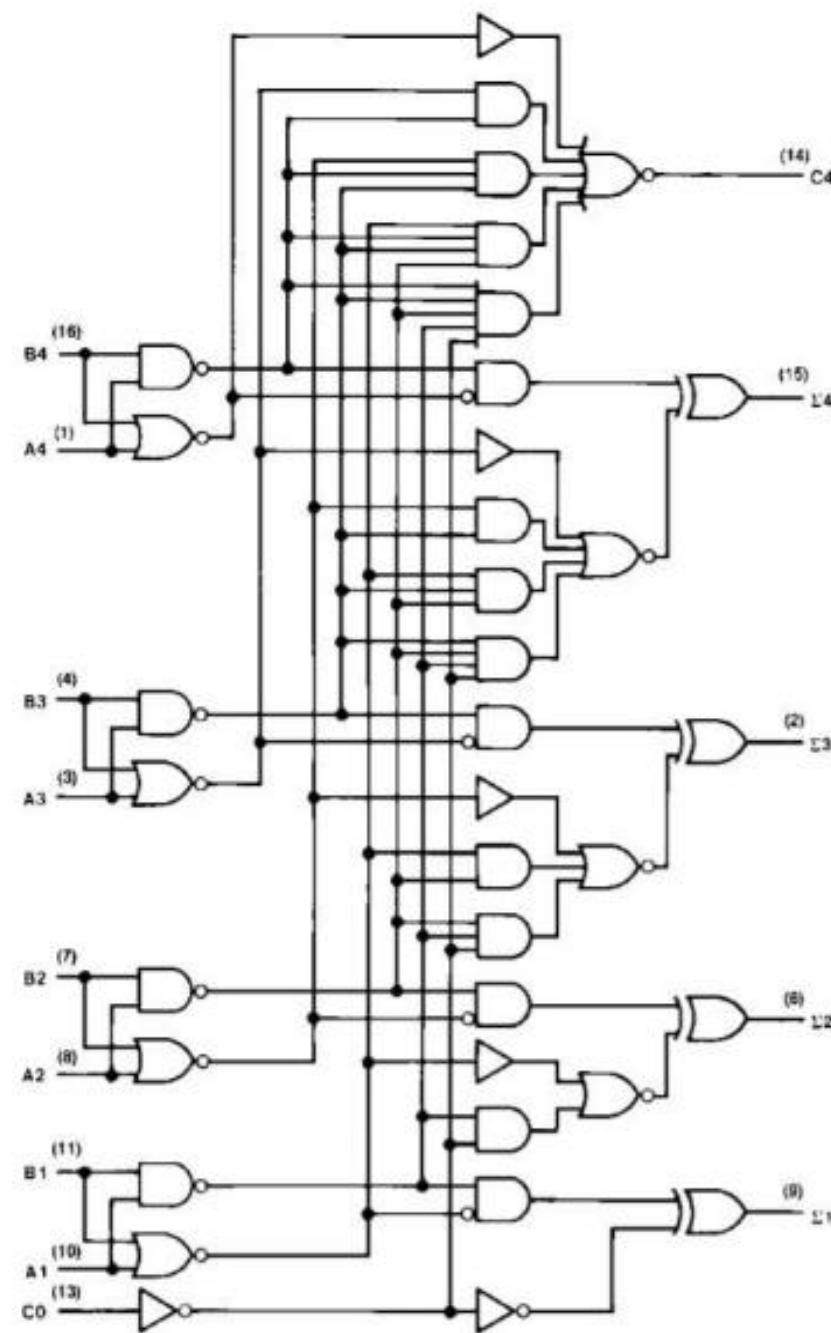
- $c_{i+3} = g_{i+2} + p_{i+2} g_{i+1} + p_{i+2} p_{i+1} g_i + p_{i+2} p_{i+1} p_i c_i$

- $c_{i+4} = g_{i+3} + p_{i+3} g_{i+2} + p_{i+3} p_{i+2} g_{i+1} + p_{i+3} p_{i+2} p_{i+1} g_i + p_{i+3} p_{i+2} p_{i+1} p_i c_i$

⇒ can compute directly from input bits and input carry c_i

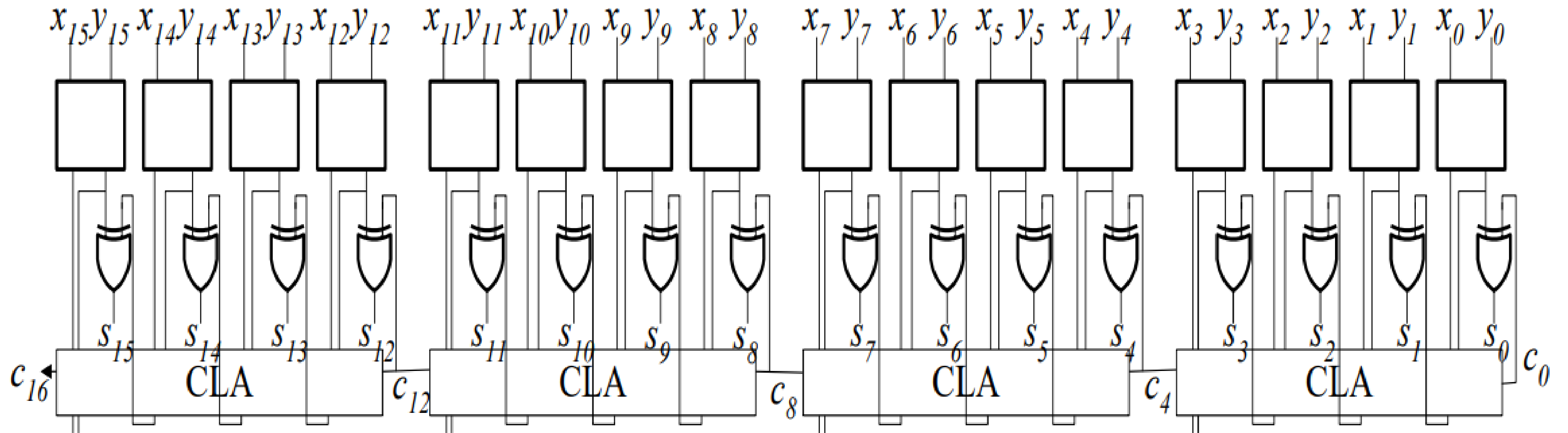
1-bit Adder에서는 c_{i+4} 를 얻기 위해 4번의 연산이 필요했던 반면, 4bit Adder은 한번의 연산으로 c_{i+1} 부터 c_{i+4} 까지의 값을 얻을 수 있으므로 연산 속도가 훨씬 빨라집니다.

4-bit Adder 회로



4 bit Binary Parallel Adder를 회로로 표현

4-bit Adder 직렬연결

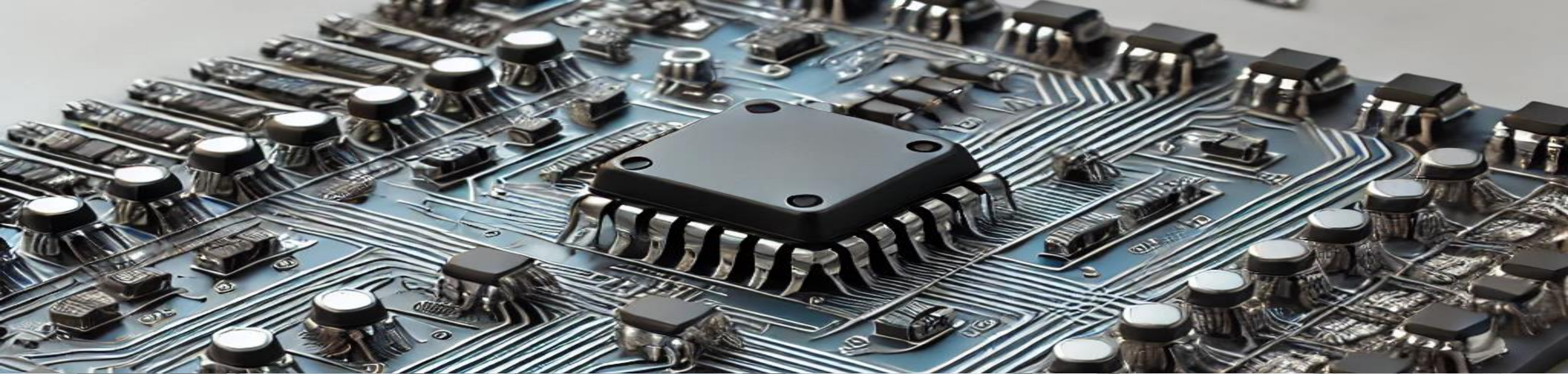


4 bit Binary Parallel Adder를 직렬로 연결하여 더욱 큰 범위의 숫자에 덧셈 연산을 할 수 있다.

Subtractor

뺄셈연산의 경우 예를 들어 4에서 3을 빼라고 할 경우 $4 - 3$ 을 $4 + (-3)$ 으로 연산한다.
양수에서 음수로 바꾸는 방법은 숫자의 2진표현에서 0을 1로 1을 0으로 바꾼 후 1을 더한다.
위 과정을 거쳐 앞서 설명한 4-bit Adder을 통해서 Subtractor 연산을 할 수 있다.

$$\begin{array}{r} 4 \quad 0100 \\ - 3 \quad -0011 \\ \hline 1 \end{array} \longrightarrow \begin{array}{r} \quad 0100 \\ + 1101 \\ \hline 1 \quad 0001 \end{array}$$



Part 2

BCD Adder /Subtractor

1-1
BCD

1-2
BCD Adder

1-3
BCD Subtractor

BCD

BCD(Binary-Coded Decimal Representation)

Decimal number의 각 숫자를 Binary number로 나타내는 방식

$$\begin{aligned} 1024_{10} &= 0000\ 0100\ 0000\ 0000_2 \\ &= 0001\ 0000\ 0010\ 0100_{BCD} \end{aligned}$$

BCD 코드와 2진수의 차이

Decimal	0	1	2	3	4	5	6	7	8	9
Binary	0000 0000	0000 0001	0000 0010	0000 0011	0000 0100	0000 0101	0000 0110	0000 0111	0000 1000	0000 1001
BCD	0000 0000	0000 0001	0000 0010	0000 0011	0000 0100	0000 0101	0000 0110	0000 0111	0000 1000	0000 1001
Decimal	10	11	12	13	14	15	16	17	18	19
Binary	0000 1010	0000 1011	0000 1100	0000 1101	0000 1110	0000 1111	0001 0000	0001 0001	0001 0010	0001 0011
BCD	0001 0000	0001 0001	0001 0010	0001 0011	0001 0100	0001 0101	0001 0110	0001 0111	0001 1000	0001 1001

BCD

계산결과가 $1001_2(9_{10})$ 이하인 경우

0000 0010
+ 0000 0100

0000 0110

Decimal	0	1	2	3	4	5	6	7	8	9
Binary	0000 0000	0000 0001	0000 0010	0000 0011	0000 0100	0000 0101	0000 0110	0000 0111	0000 1000	0000 1001
BCD	0000 0000	0000 0001	0000 0010	0000 0011	0000 0100	0000 0101	0000 0110	0000 0111	0000 1000	0000 1001
Decimal	10	11	12	13	14	15	16	17	18	19
Binary	0000 1010	0000 1011	0000 1100	0000 1101	0000 1110	0000 1111	0001 0000	0001 0001	0001 0010	0001 0011
BCD	0001 0000	0001 0001	0001 0010	0001 0011	0001 0100	0001 0101	0001 0110	0001 0111	0001 1000	0001 1001

BCD

계산결과가 $1010_2(10_{10})$ 이상인 경우

0000 1000
+ 0000 0110

0000 1110

Decimal	0	1	2	3	4	5	6	7	8	9
Binary	0000 0000	0000 0001	0000 0010	0000 0011	0000 0100	0000 0101	0000 0110	0000 0111	0000 1000	0000 1001
BCD	0000 0000	0000 0001	0000 0010	0000 0011	0000 0100	0000 0101	0000 0110	0000 0111	0000 1000	0000 1001
Decimal	10	11	12	13	14	15	16	17	18	19
Binary	0000 1010	0000 1011	0000 1100	0000 1101	0000 1110	0000 1111	0001 0000	0001 0001	0001 0010	0001 0011
BCD	0001 0000	0001 0001	0001 0010	0001 0011	0001 0100	0001 0101	0001 0110	0001 0111	0001 1000	0001 1001

Different!

BCD Adder

계산결과가 $1001_2(9_{10})$ 이하인 경우
 $+0000_2(0_{10})$

계산결과가 $1010_2(10_{10})$ 이상인 경우
 $+0110_2(6_{10})$

0000 1000
+ 0000 0110

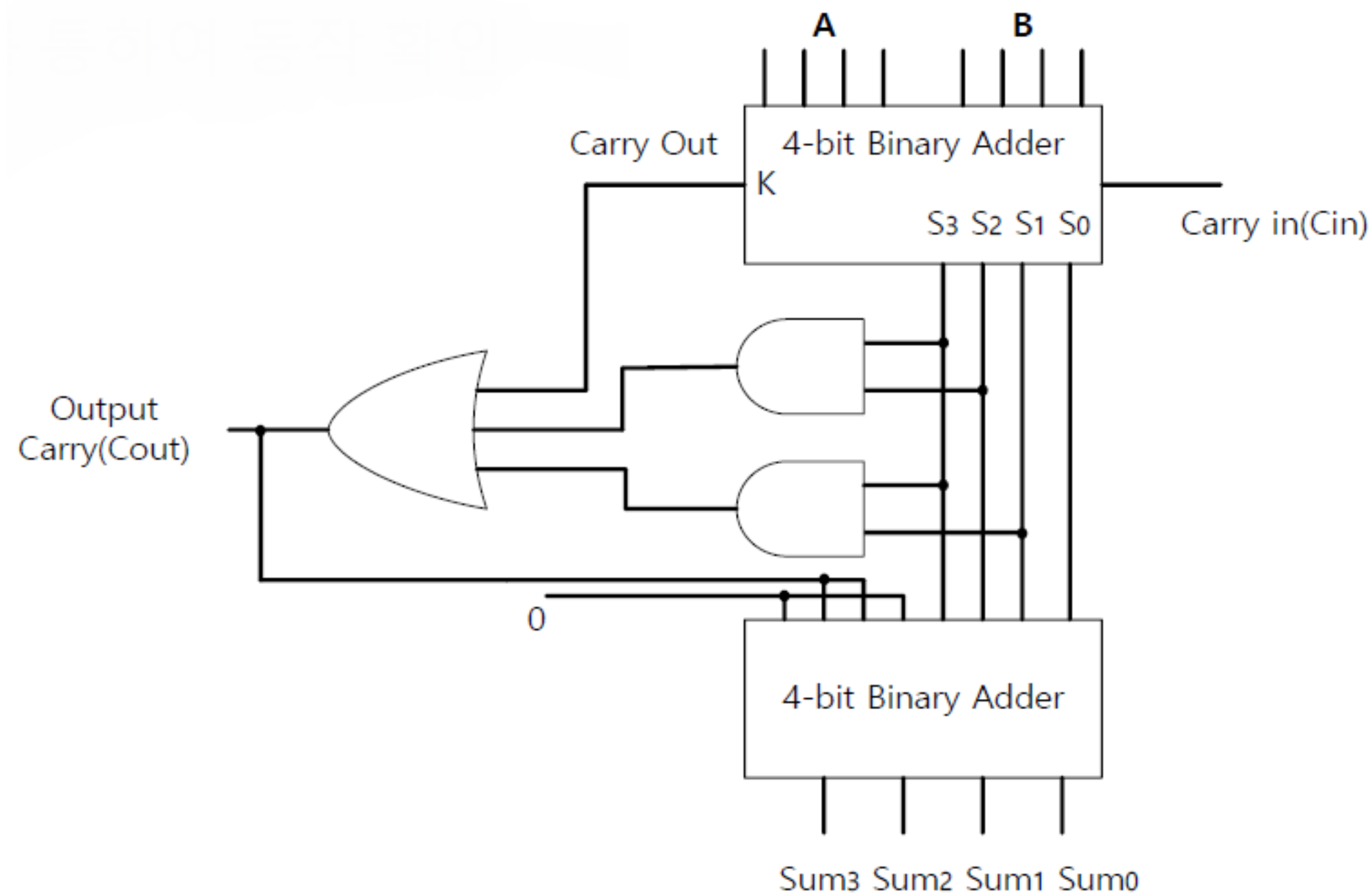
0000 1110(2)

0000 1110
+ 0000 0110

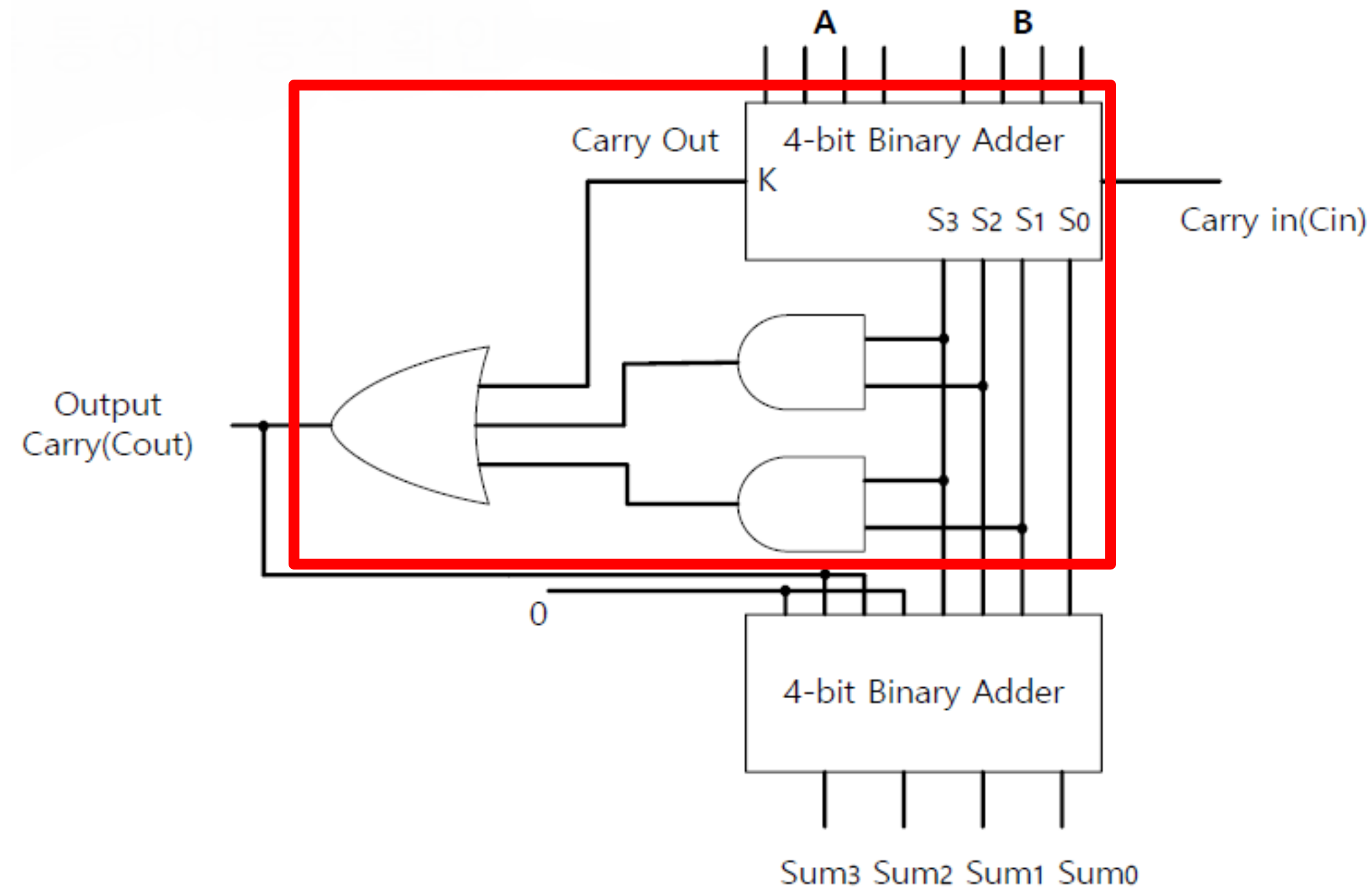
0001 0100(BCD)

BCD Adder

BCD Adder 두 BCD 코드를 더하는 가산기



BCD Adder



계산결과가 $1001_2(9_{10})$ 이하인 경우
 $+0000_2(0_{10})$

계산결과가 $1010_2(10_{10})$ 이상인 경우
 $+0110_2(6_{10})$

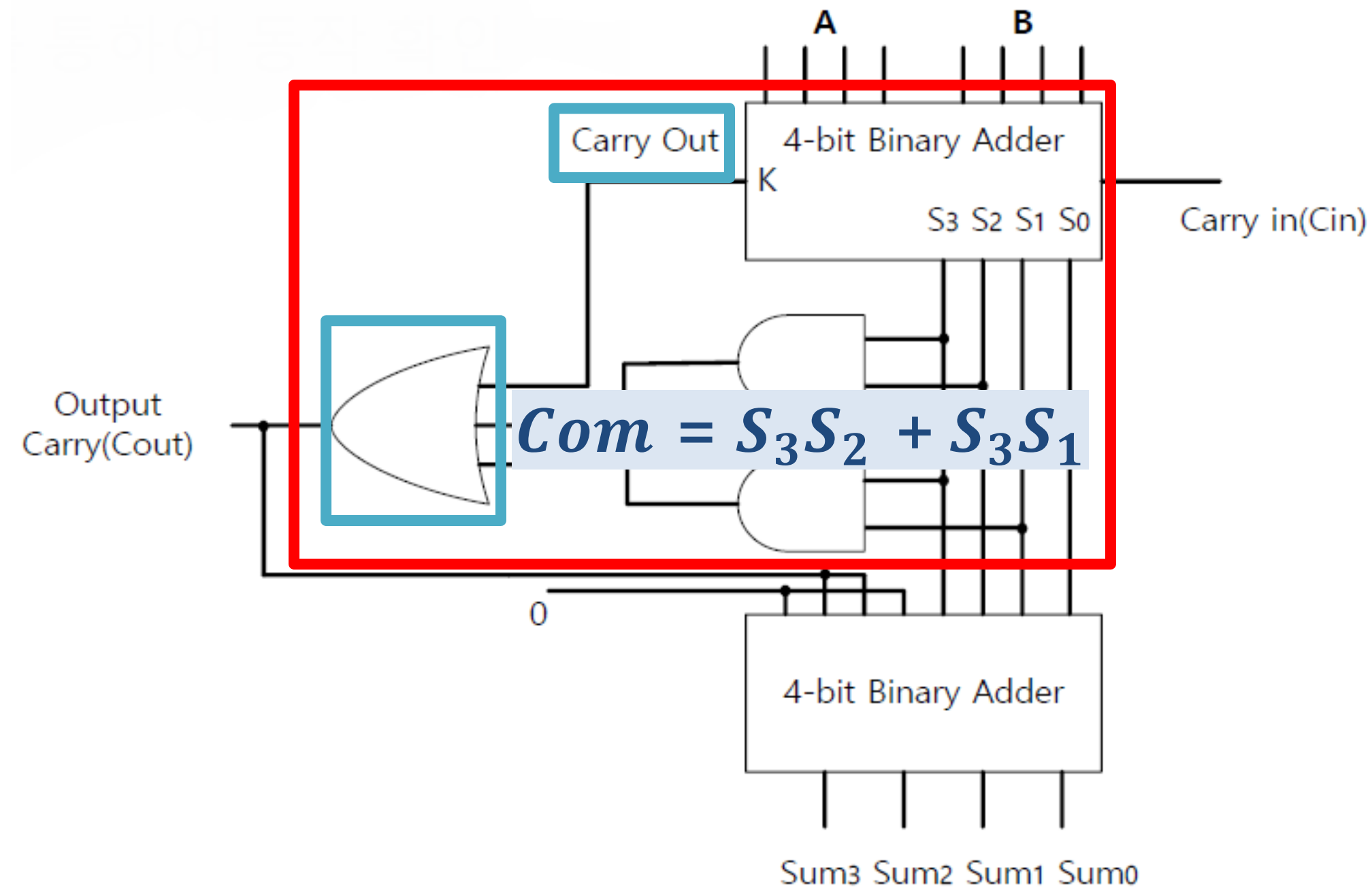
BCD Adder

S_3	S_2	S_1	S_0	Com
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

Com	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	1	1	1	1
10	0	0	1	1

$$Com = S_3S_2 + S_3S_1$$

BCD Adder

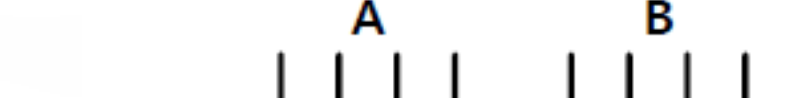


계산결과가 1001_2 이하인 경우
 $+0000_2$

계산결과가 1010_2 이상인 경우
 $+0110_2$

U

계산결과가 10000_2 이상인 경우
 $+0110_2$

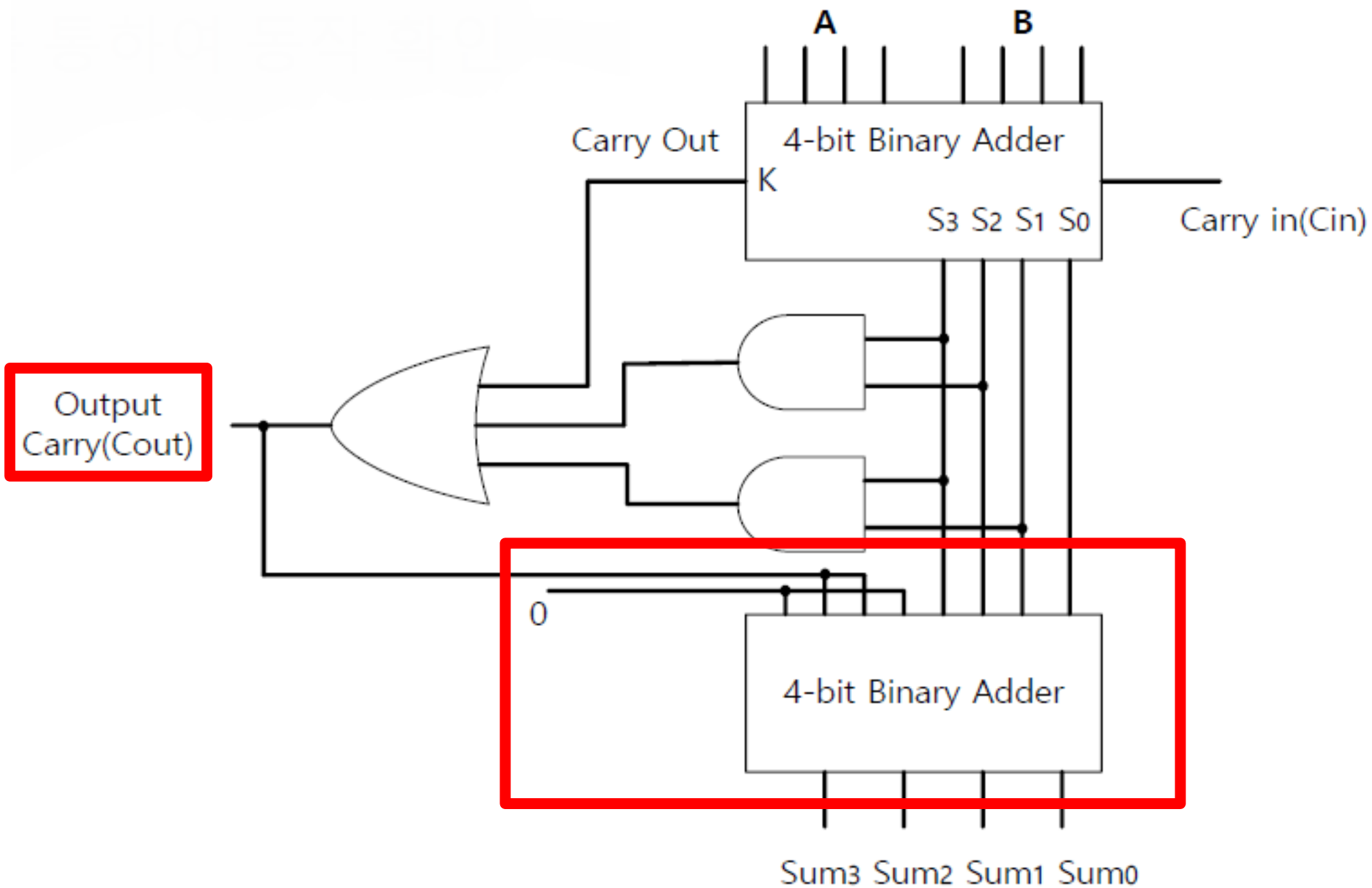


$$\begin{array}{r} 0000\ 1000_2 \\ + 0000\ 1001_2 \\ \hline 0001\ 0001_2 \end{array}$$

$$\begin{aligned} 0001\ 0001_2 &= 17_{10} \\ 0001\ 0001_{BCD} &= 11_{10} \end{aligned}$$

계산결과가 1010_2 이상인 경우 + 0110_2

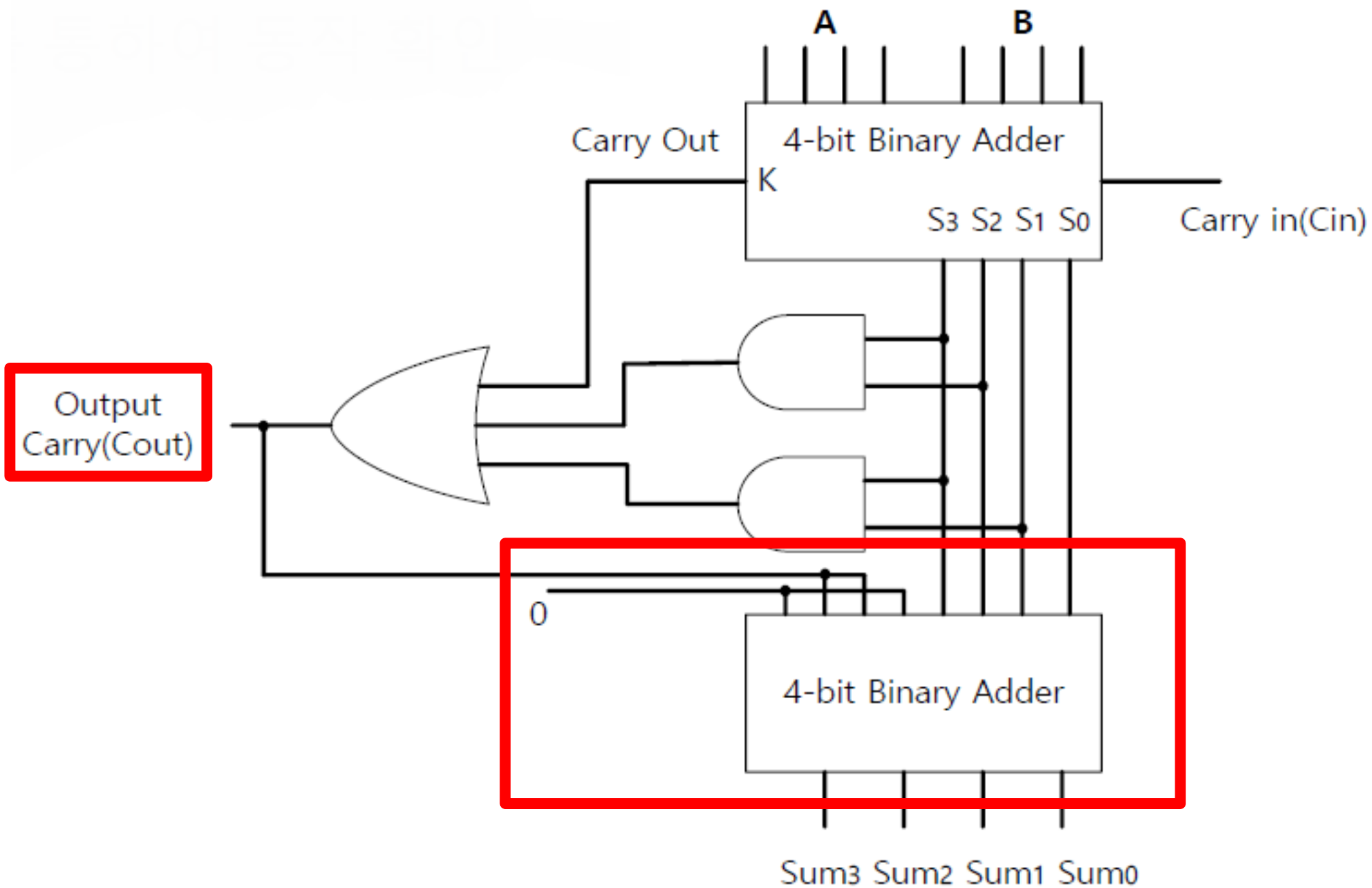
BCD Adder



계산결과가 $1010_2 (10_{10})$ 이상인 경우
 $0110_2 + S_3S_2S_1S_0_2$

계산결과가 $1001_2 (9_{10})$ 이하인 경우
 $S_3S_2S_1S_0_2$
($\because 0000_2 + S_3S_2S_1S_0_2$)

BCD Adder



$$1000_{BCD} + 1001_{BCD} = 8_{10} + 9_{10}$$

$$\begin{array}{r} 0000\ 1000_2 \\ + 0000\ 1001_2 \\ \hline \end{array}$$

$$\begin{array}{r} 0001\ 0001_2 \\ + 0000\ 0110_2 \\ \hline \end{array}$$

$$0001\ 0111_2$$

$$0001\ 0001_2 = 17_{10}$$

$$0001\ 0111_{BCD} = 17_{10}$$

BCD Subtractor

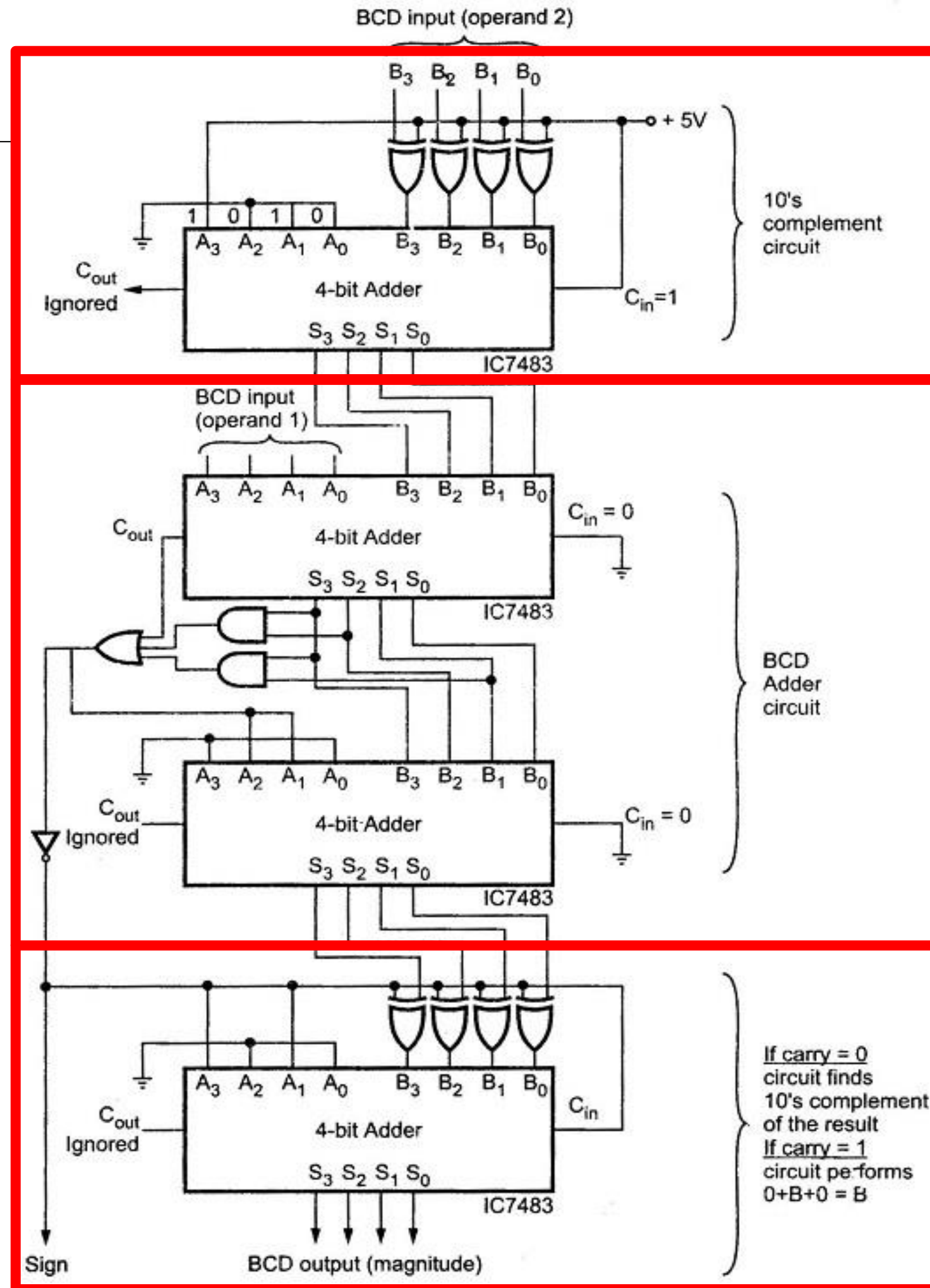
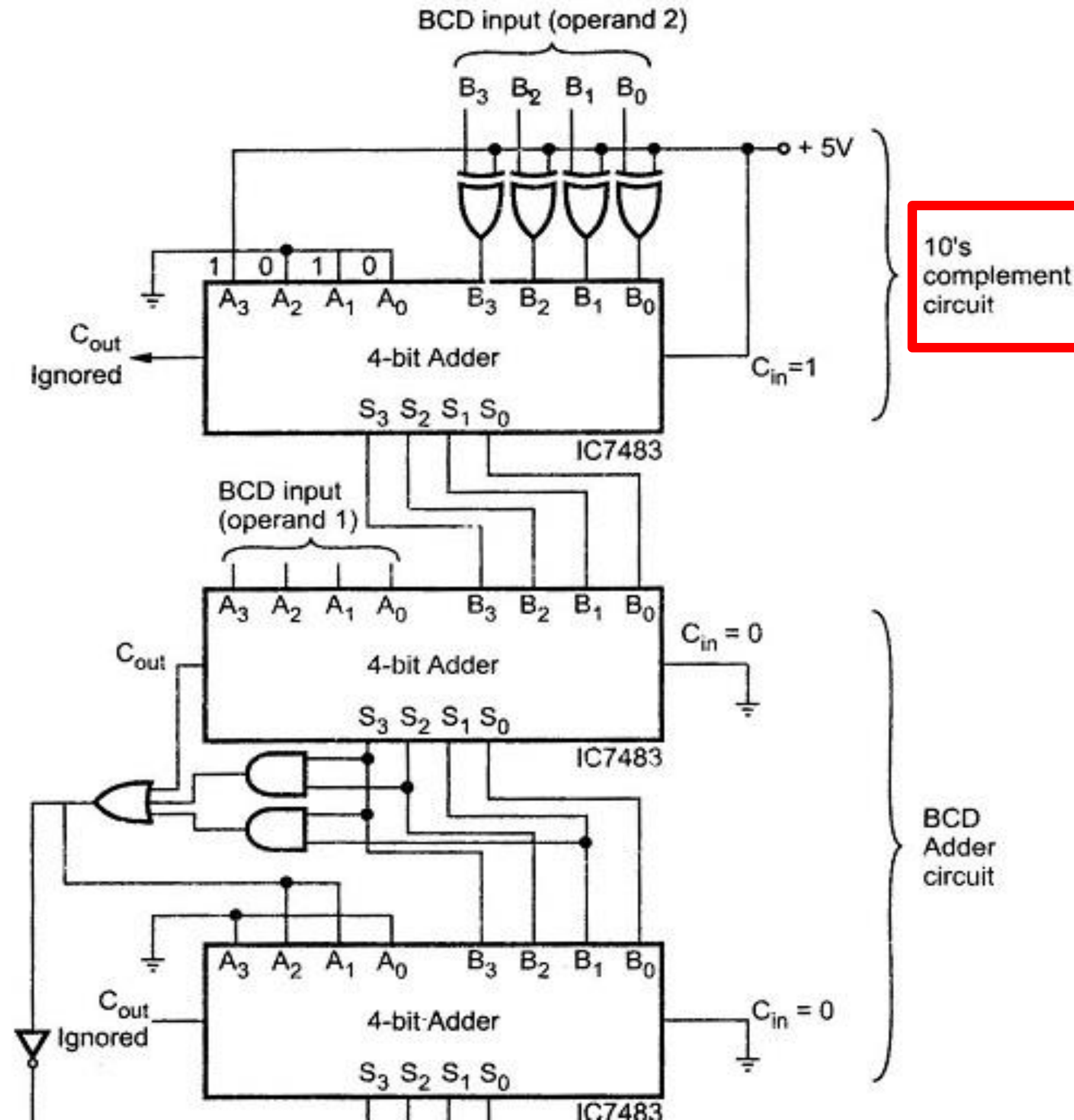


Fig. 3.35 4-bit BCD subtractor using 10's complement method

BCD Subtractor



10's complement

Decimal number의
9's complement를 구한 후 +1

$$6_{10}$$

$$\begin{aligned} \text{9's complement} &: 9_{10} - 6_{10} = 3_{10} \\ \text{10's complement} &: 3_{10} + 1_{10} = 4_{10} \end{aligned}$$

A = 1010₂



10's complement

$$B_3 = B_3 \oplus 1$$

$$B_2 = B_2 \oplus 1$$

$$B_1 = B_1 \oplus 1$$

$$B_0 = B_0 \oplus 1$$

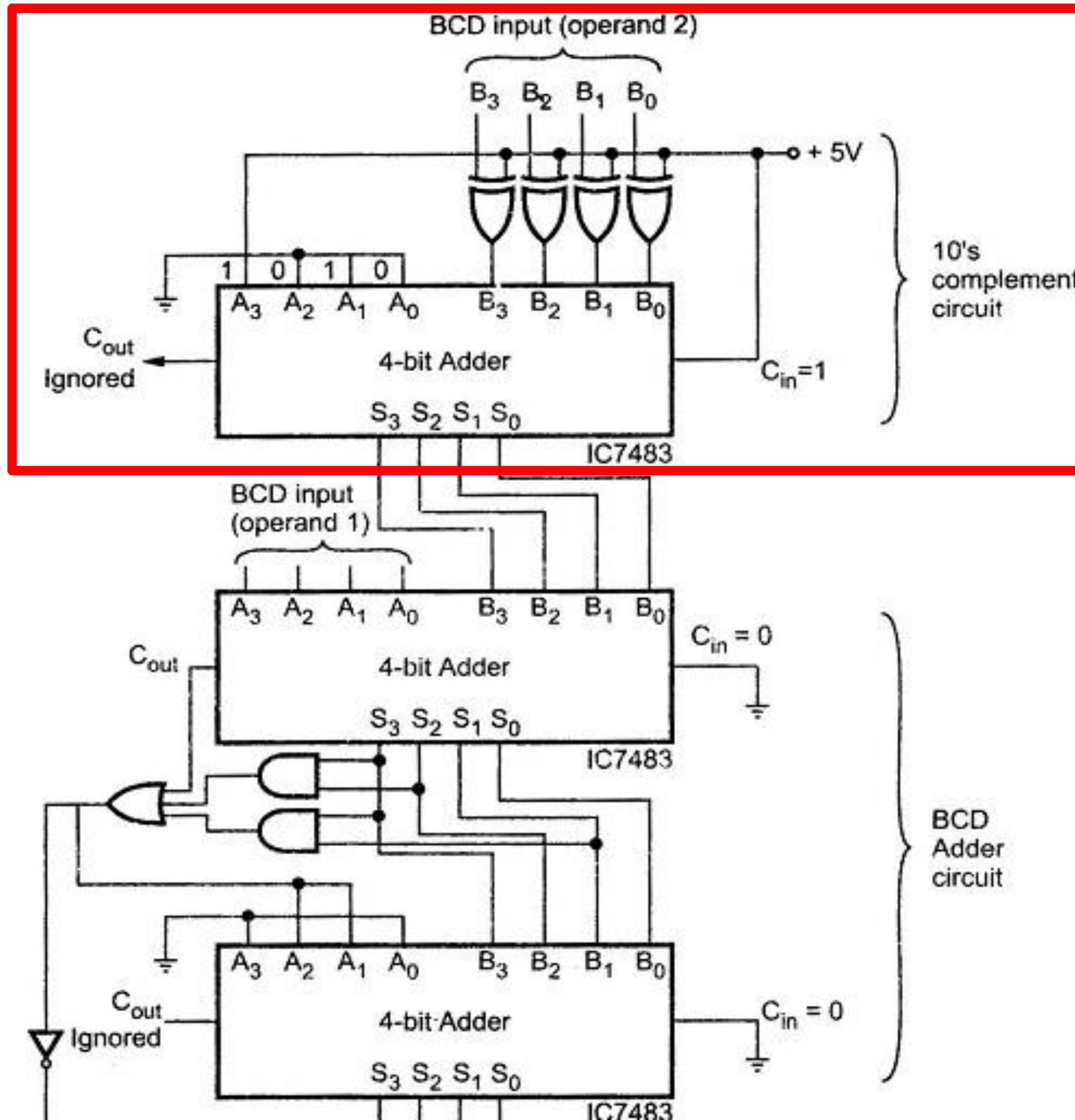
$$A_3 = 1$$

$$A_2 = 0$$

$$A_1 = 1$$

$$A_0 = 0$$

BCD Subtractor



10's complement

$$A = 0011_2(3_{10})$$

$$B = 0100_2(4_{10})$$

1 111

1010₂

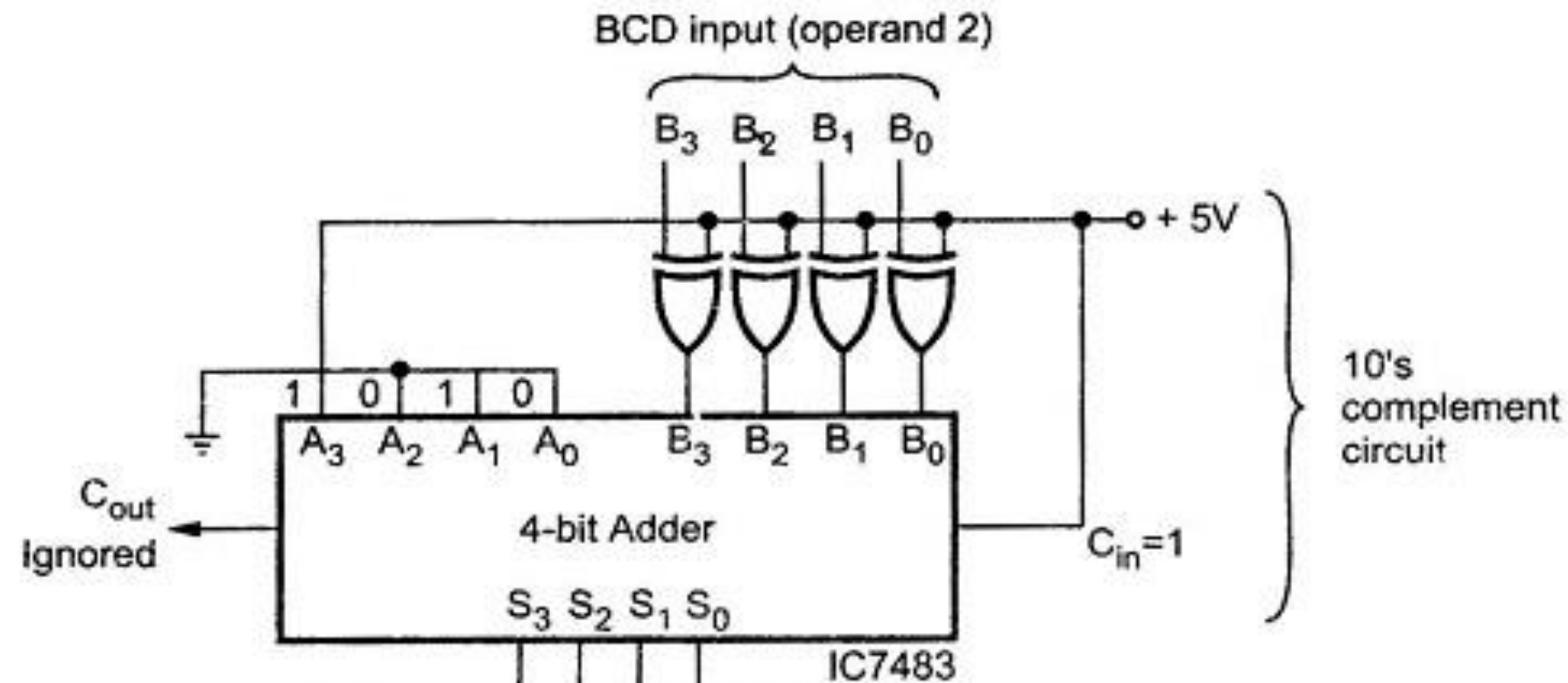
+ 1011₂

1 0110₂

Ignore carry

0110₂

BCD Subtractor



$$A = 0011_2(3_{10})$$

11

$$\begin{array}{r} 0011_2 \\ + 0110_2 \\ \hline \end{array}$$

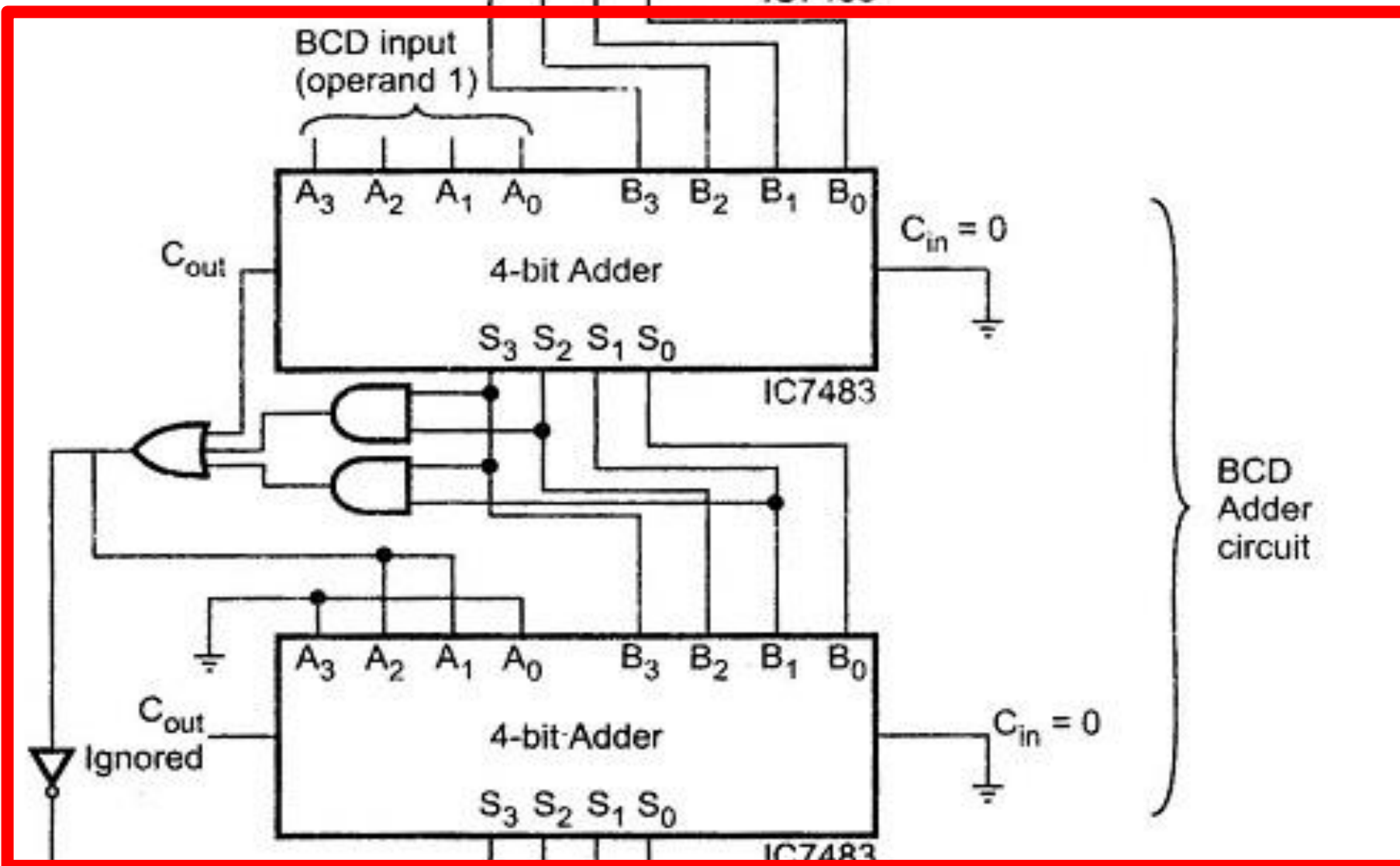
$$1001_2$$

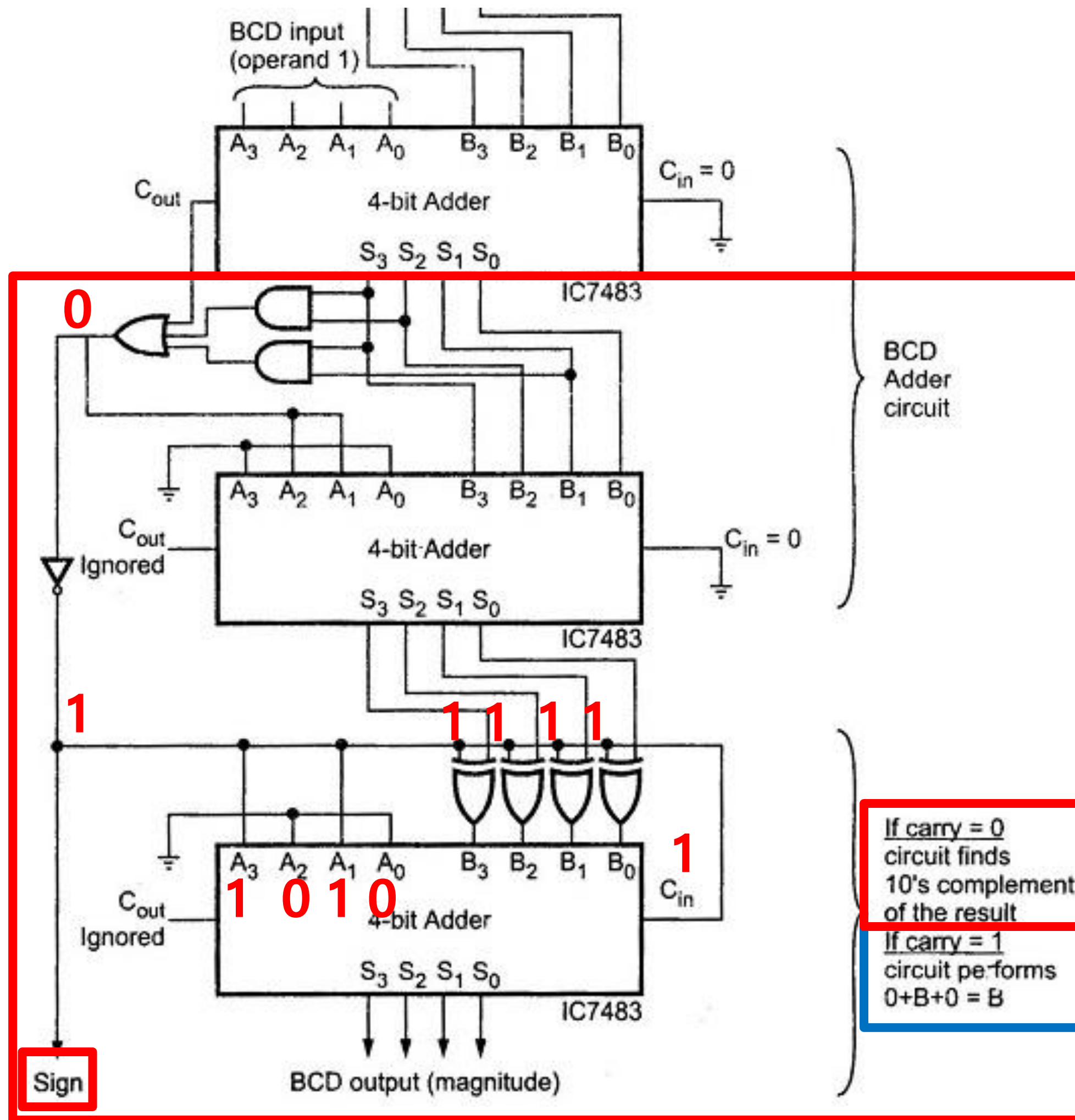
$$1001_2(< 1010_2)$$

$$\begin{array}{r} 1001_2 \\ + 0000_2 \\ \hline \end{array}$$

$$1001_2$$

$$1001_2 \neq -1_{10}$$





$$\begin{array}{r} 1001_2 \\ \text{XOR } 1(\text{반전}) \\ 0110_2 \end{array}$$

$$\begin{array}{r} 1111 \\ 1010_2 \\ + 0110_2 \end{array}$$

$$10001_2$$

$$\begin{array}{r} \text{Ignore carry} \\ 0001_2 \end{array}$$

$$-0001_{BCD} = -1_{10}$$

출처 및 기여도

김주호 교수님 디지털회로개론 강의자료
컴퓨터공학실험II 10주차 강의자료
<https://www.eeeguide.com/bcd-subtraction/>

박정수 : Part 1 발표자료 제작, 발표(50%)
이동협 : Part 2 발표자료 제작, 발표(50%)